

Simple TWO-WAY INTERCOM

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If you have to frequently get up to talk to someone in your place, this intercom can save you the trouble. You can just switch it on from both sides to communicate. It can also come in handy if you have someone observing isolation at home due to Corona. Block diagram of the intercom is shown in Fig. 1.

Circuit diagram of the intercom

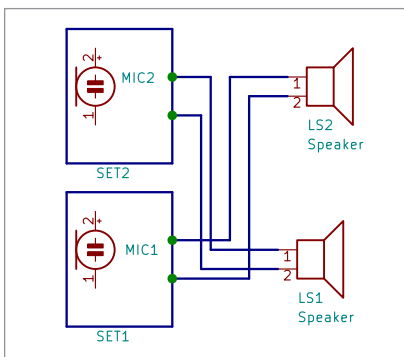


Fig. 1: Block diagram

is shown in Fig. 2. The simple, low-cost intercom is built around two condenser microphones (MIC1 and MIC2), four 2N2222 transistors (T1 through T4), and two LM380N audio amplifiers (IC1 and IC2).

A condenser microphone is like a capacitor (also called condenser) with two parallel plates placed close to each other. One of the plates is made of a very light and thin material that acts as the diaphragm. The diaphragm vibrates when struck by sound waves, changing the distance between the two plates, and thereby varying the capacitance. The resulting fluctuation in capacitance creates the corresponding electrical signal, which is amplified.

Most intercoms do not use a pre-amplifier circuit, because of which you get distorted sound output. This circuit employs pre-amplifiers to en-

hance the inputs for loud and clear sound from the loudspeakers.

As shown in Fig. 2, there are two identical circuits. In the lower circuit, R1 works as a current-limiting resistor for MIC1. The electrical signal produced by sound flows through coupling capacitor C1, which blocks any DC signal and allows only AC signals to pass through. This signal is boosted by the amplifier sections formed by transistors T1 and T2 and IC LM380N.

The signal is first boosted by the amplifier built around transistor T1 and further by the second amplifier built around transistor T2. The output at collector of T2 goes to LM380N (IC1) through its pin 2, via capacitor C5, and the pot VR1.

The 8-ohm speaker (LS2) connected to output of

PARTS LIST

Semiconductors:

IC1-IC2 - LM380N audio amplifier

T1-T4 - 2N2222 NPN transistor

Resistors (all 1/4-watt, $\pm 5\%$ carbon), unless stated otherwise:

R1, R2 - 10-kilo-ohm

R3, R4, R7, R8 - 100-kilo-ohm

R5, R6, R9, R10 - 1-kilo-ohm

R11, R12 - 18-ohm

VR1, VR2 - 100-kilo-ohm potmeter

Capacitors:

C1-C4 - 10 μ F, 25V electrolytic

C5, C6 - 47 μ F, 25V electrolytic

C7, C8 - 100 μ F, 25V electrolytic

Miscellaneous:

S1, S2 - On/off switch

MIC1, MIC2 - Electret condenser microphone

BATT.1, BATT.2 - 9V battery

LS1, LS2 - 8-ohm, 1-watt speaker

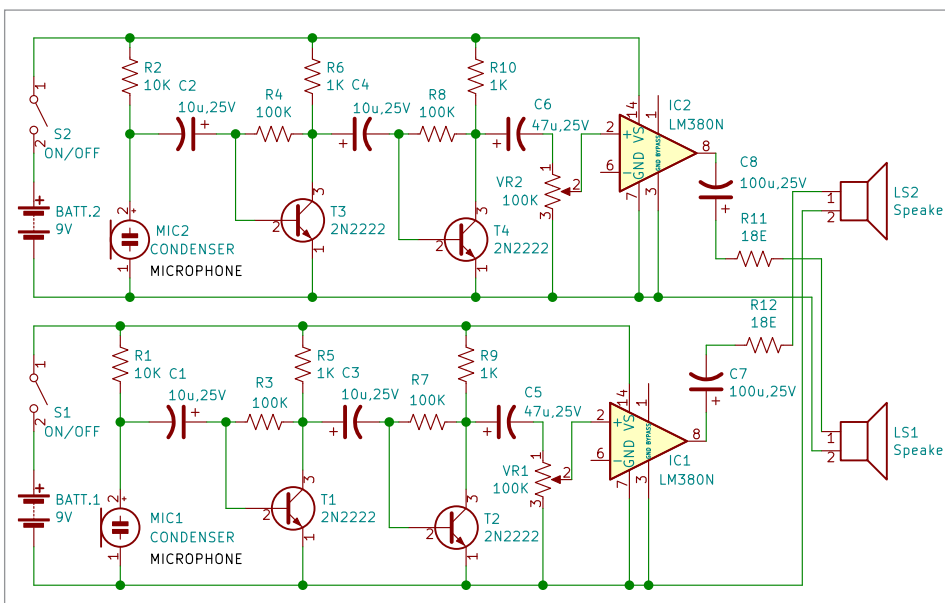


Fig. 2: Circuit diagram